

DevOps COURSE

While students in large colleges study massive amounts of theory, we offer up to date, fresh and relevant DevOps classes **focused on practical work methods, adapted to industry needs** so you can penetrate the job market with enough confidence and the right experience to do your job right.

Our classes are taught by industry experts, those who work simultaneously as interviewers and recruiters in high-tech companies and know exactly what it takes to succeed. Each student learns **exactly** what they need to know for their future jobs – for this reason, all candidates are screened and evaluated before admission in order to guarantee the highest level of learning and ensure future career opportunities.

What does this mean for you? You gain the best hands-on experience and pay less money - two birds, one stone.

Our knowledge, your future



Private classes

Our DevOps courses focus on practical knowledge; in class exercises, homework assignments and learning in small groups which allows for personal attention and better understanding of the material.



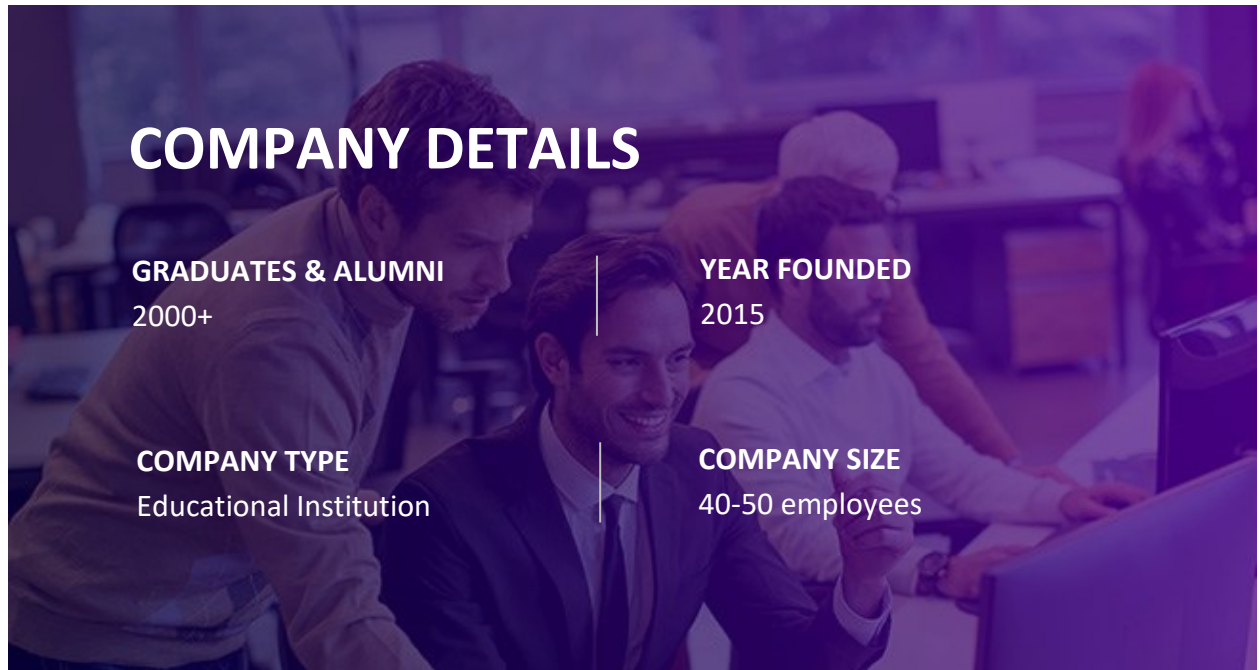
Classes for companies

We offer customized DevOps courses and workshops according to your company needs. Course materials are suited to your everyday tasks and training requirements.



“Preparation for Work” workshop

We can provide career assistance by reviewing your resume, teaching social media networking and defining LinkedIn content for professional “branding” as well as refer you to relevant positions.



About the program

In today's dynamic tech world, DevOps stands out as a game-changer.

It's the bridge that connects software development and IT operations in today's cloud-based environments, ensuring rapid software delivery, robust infrastructure, and top-notch quality.

At its core, DevOps is about collaboration. It's about tearing down traditional barriers and working together to innovate faster, adapt to market changes swiftly, and elevate the overall tech experience.

Our training program is your gateway to diving deep into this transformative role. Crafted by seasoned DevOps professionals from leading tech companies, this program provides a blend of real-world insights and hands-on practice. These aren't instructors who simply know the theory; they live the DevOps life daily, which means what you learn is relevant, practical, and empowering.

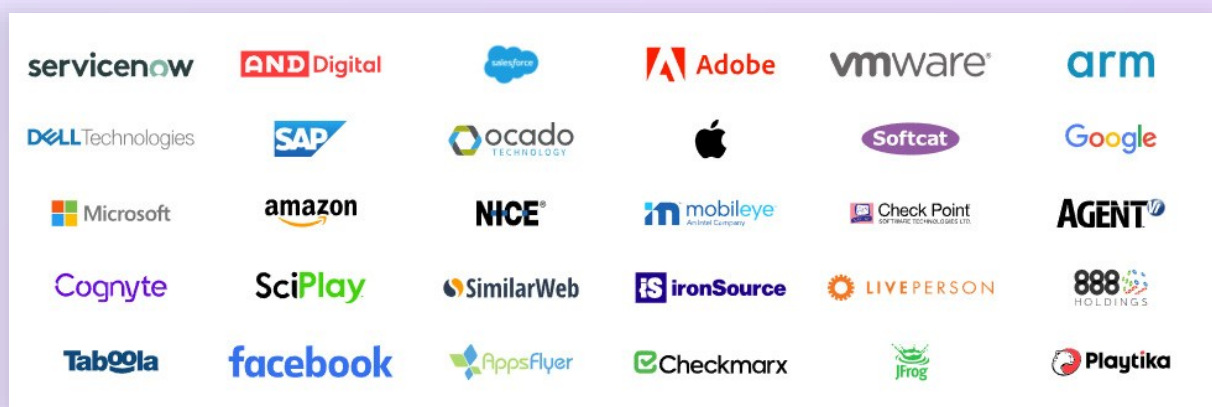
Joining this program is more than just a learning experience; it's a step towards a promising career transformation. We're here not only to equip you with DevOps knowledge but also to mentor and guide you as you transition into this high-demand role. Dive in, be inspired, and let us help you become a pivotal force in the tech industry.

Who is this course for?

DevOps career opportunities are on a steep rise worldwide. Organizations are investing heavily in DevOps capabilities to maintain a cutting edge in the market. Our DevOps training program will benefit professionals with the following backgrounds:

- IT engineers
- Software Developers
- Architects
- Technical Project Managers
- Operations Support
- Deployment engineers
- Dev managers
- Automation testers
- Decision makers
- Those with a technical background who are interested in learning DevOps skills at the level required by successful, leading companies

OUR ALUMNI WORK WITH THE BEST





Learn from industry experts

Industry-recognized DevOps engineer program will teach you current and in-demand skills, ensuring you stay ahead of the curve in a fast-changing industry.



Get hands-on experience

Practical skills are key to stand out in the market. By working on practical tasks throughout the program, you'll master the skills of a great DevOps professional.



Learn amongst professionals

You are a who you spend time with. Studying alongside high-level peers with a similar goal gives you added insight and a solid professional network.



Connect with the industry

Our Career Mentorship program provides dedicated support from high level recruiters from leading companies, ensuring your DevOps career success.

THE INSTRUCTORS



Danny Gitelman
Senior Site Reliability Engineer
Microsoft



Eduard Brook
DevOps Team Lead
Walmart Global Tech



Daniel Gotlieb
DevOps Team Lead
Trigo



Doron Nuni
R&D Group Manager
Imperva



Yarin Galmor
Senior DevOps Engineer
Palo Alto Networks



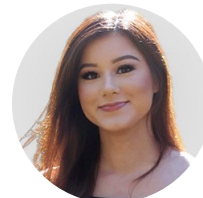
CAREER BRANDING WORKSHOP



Adi Barkai
Employee Experience Leader
Amazon



Dganit Feller
Director Talent Acquisition
Pentera



Stephanie Dang
Senior Talent Acquisition
NCR Voyix



DEVOPS COURSE SYLLABUS

Scripting for DevOps Engineers

Engineers automating infrastructure and managing configuration need to know how to code. In this module, you will get a hands-on introduction to the Python programming language as well as advanced techniques that will ease your Development learning process.

Build Automation and Continuous Integration

An effective DevOps leverages technology to power continuous integration and a continuous delivery pipeline. In this module, you will learn how to configure Jenkins to run pipelines, code coverage and quality tools, testing suites, and CM and deployment tools.

Version Control Using Git

In this module, you will learn why and how to use a version control system for DevOps. You will use Git to version, manage, merge, diff, and control your infrastructure code.

Configuration Management

Automating configuration management tasks helps gain speed, agility, and productivity. In this module, you will learn how to use technology to automate these tasks.

Provisioning Resources

In this module you will learn how to provision resources across over 20 different on-premises and cloud platforms and how to create and provision resources in a hybrid infrastructure.

Working with Containers

In a DevOps program, Docker containers can be used to simplify deployments. In this module, you will learn how to build containers and compose multi-container applications to support microservices.

Linux Fundamentals

Once you've established a DevOps program, you will learn how to setup, configure, administer, and manage data center and Cloud-based environments. In this module, you will learn the Linux skills required to build these foundations.

Data and Continuous monitoring

In this module, you discover how to collect, analyze, and make decisions using logs and other system-generated data. You ingest and analyze log and other system data to provide operational troubleshooting and decision support capabilities.

<i>Topic</i>	<i>Description</i>
Introduction to DevOps Concepts	
Introduction	• What is DevOps? • Methodologies • The role of ops in the DevOps world
Scripting for DevOps Engineers	
Overview and Setup	• Intro • Python environment setup • IDE setup
Data and Control Flow	• Data types • Syntax • Operators • Statements • Conditions • Loops
Data Structures	• Array • List • Tuple • Dictionary
Advance Python	• Files I/O • Error handling • Debug • Packages • PIP • Conventions
Web Fundamentals	• Introduction • Elements • Attributes • CSS • JavaScript

Test Automation	• Web drivers • Methods • Locators • Controllers • Switch and Navigation • Synchronization
API's	• Overview • Protocols • Why REST • Requests API • Rest methods • Response codes • Structure • JSON rules • JSON parsing • Routing • Path params • Query params • Building an API
Python Project	• Build a python RESTful API
<i>Build Automation and Continuous Integration</i>	
CI Intro	• Overview • Jenkins Introduction • Exploring Jenkins dashboard • Jenkins Architecture
Jenkins Core	• Setup & configuration • Jenkins configurations • Pipelines • Blue Ocean
Build Jobs	• What are build Jobs • Building your build jobs • Scheduling • Reporting • Disabling and deleting jobs • Post build actions • Pipelines as code

Jenkins Management	• Plugins • Log rotation • Master-slave architecture • Nodes restrictions • Reporting
Jenkins Security	• Authentication • Authorization • Creating users • Jenkins API
<i>Version Control Using Git</i>	
Git Basics	• What is Git? • Git VS. SVN • Terminology • Setup & configuration • Configuring Git • Logging • Commits • Branching • Merging • Conflicts • Reset & Revert • Git GUI
Git & Repository	• Repository overview • Remotes • Cloning • Fetch, pull & push • Building your repository • Pull Requests • Git – Jenkins integration

Data and Continuous monitoring

Database

- Overview and setup
- RDBMS Vs. NoSQL
- Types
- Connections
- Security
- Schemas and tables
- Columns properties
- Data types
- PyMySQL
- Cursor
- DB (CRUD) operations

Data Monitoring

- Monitoring Concepts
- Introduction to Prometheus stack
- Prometheus server
- Metrics collection
- PromQL
- Exporters
- Grafana
- Alert Manager

Linux Fundamentals

Overview

- What is Linux?
- Distributions
- Environment setup
- Oracle VirtualBox & VMware

Linux Commands

- Shell, Bash & Terminal
- Commands
- Input / Output
- Package management
- Interprocess communication
- Grep, tail, sed

Linux System	• System Utilities • Linux processes • System signals • VI & VIM & nano • SSH server & client • Public and private keys • SFTP
Working with Containers	
Overview	• What is, and why Docker? • Use case of Docker • Container Life Cycle • Docker vs. Virtualization • Docker types • Environment setup • Docker commands
Docker Basic	• Docker architecture • YAML • Docker images • Docker compose • Services • Docker image download • Working multiple containers • Contexts • Docker HUB
Docker Advanced	• VNC (Virtual Network Computing) • Docker-Jenkins integration • “Dockerizing” your code • Creating a custom image • Running a container from the custom image • Docker networking • Docker volumes • Best practices • Docker-Jenkins integration • Containers dependencies • Environment files and configurations

Container Orchestration - Kubernetes	• Intro to Kubernetes • Kubernetes Architecture • Environment Setup • Kubernetes Components Hierarchy • The K8S API
Working with Kubernetes	• Objects • API versions • Desire and actual state • Pods, ReplicaSets, Deployment • Updates and reverts • Scaling and auto scaling • Services, networking options, Exposing our cluster • Config maps • Secretes • Kubectl • Logging
HELM	• Intro to Helm • Architecture • Developing and managing helm charts • Templating • Parameterized deployments • Debugging • Updating charts • Requiring dependency as charts • Sharing chart with repositories
Hosted Kubernetes	• Container Services in the cloud (GCP, AWS) • K8S On premise
CI/CD Project	• Build and ship your containerized python application platform using CI/CD pipelines

<i>Provisioning Resources</i>	
AWS Hands-On	• AWS CLI • Cloud app deployment • Connecting to cloud remote machines • Working with access key and secret access keys • Profiles • Budgets • Cost explorer
Terraform	• Infrastructure as code • Environment setup • HCL • Workflows • Commands • Providers • Resources • Variables • User data • Output values • Cloud infrastructure
Cloud Platform	• Overview • Why cloud? • GCP Vs. AWS Vs. Azure • Cloud deployment
AWS (Amazon Web Services) Basics	• AWS account • Regions • EC2 types and pricing • AMI's • EIP, allocating, associating, releasing • Launch instances in AWS • Identity Access Management (IAM) • Roles • Policies • Users • Groups

<i>Configuration Management</i>	
Configuration Management	• Introduction • Terminology • Modules • Configurations management tools • Master-Agent configurations • Agentless operation • Roles • Environments
Ansible	• Introduction to Ansible • Ansible advantages • Ansible Installation • Provisioning • Built-in security • Configuring Ansible Roles • Write Playbooks • Executing Ad-Hoc command • Declarative commands • Facts • Modules • Variables • Ansible galaxy • Conditionals • Loops • Vault
<i>Final Project</i>	
Final Project	Orchestrating your containerized deployments for production-ready products
<i>Job Interviews</i>	
Job Interviews	• Most frequent interview questions and solutions • Course summary

Training Program Project

To keep this training ultra-practical, we've gone beyond the weekly assignments that let you practice each new tool, technology, and method you learn.

We have designed a 4-part, hands-on project that allows you to experience how all the pieces you learn fit together, simulating a real-world environment, and preparing you for work.

We highly recommend completing the project at a high level so you can proudly showcase it in your github to demonstrate your skills in a tangible way.

Part 1

You will design, build, document, and test a web-based Python frontend and backend stack with an underlying DB.

Part 2

You will create an industry-level CI/CD pipeline for your Python product.

Part 3

You will dockerize your app using best practices for building, testing and optimizing your Docker images.

Part 4

You will deploy your highly scalable Python product using CI/CD pipelines achieving a production-level product.

What You Get!

When we started our training programs, we set out to create the most effective learning experience for our students. After a good amount of trial and error, we've finally nailed it. Here's what you get as our student:

- ✔ Digital copies of 200+ pages of DevOps course notes
- ✔ Access to 15 live, 3-hour online lessons from the best minds in DevOps today as well as 24-month access to lesson recordings
- ✔ Access to a private WhatsApp group with your instructor and your classmates for close guidance and feedback
- ✔ 12 hours of our Career Branding Workshop, led by senior HR and recruitment professionals in Meta and Cybereason
- ✔ Exclusive invitation to join our Marketplace where your next-level job might be waiting for you
- ✔ All course worksheets, links, and other course materials.